NFA to DFA Converter

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Handling State 1

1. Calculate the ϵ-closure of the state in M; call the result S. From the ϵ-closure table, we know that $E(\{1\}) = \{1, 3\}$.

2. Calculate the transitions in M from each state in R on each input character.

   a Here we must collect all places we can get to when reading an **a** starting in any state in $\{1, 3\}$. Since

   - $\delta(1,$ **a** $) = \{2\}$
   - $\delta(3,$ **a** $) = \{5\}$

   we see then that $\delta_R(1,$ **a** $)$ includes at least $\{2, 5\}$.

   b Here we must collect all places we can get to when reading an **b** starting in any state in $\{1, 3\}$. Since

   - $\delta(1,$ **b** $) = \varnothing$
   - $\delta(3,$ **b** $) = \{1, 2, 5\}$

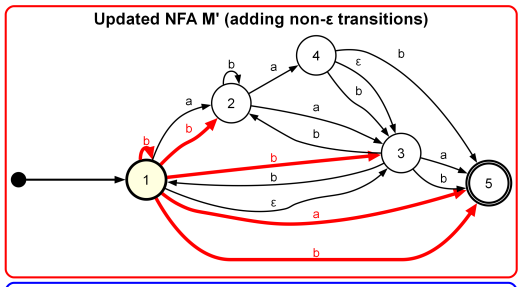
   we see then that $\delta_R(1,$ **b** $)$ includes at least $\{1, 2, 5\}$.

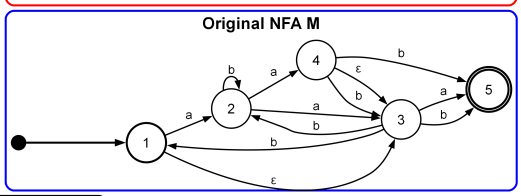
3. For each input character, calculate the ϵ-closure of the resulting set of states.

   a Merging the relevant rows in the table for $E(\{2, 5\})$, we see that $\delta_R(1, a) = \{2, 5\}$.

   b Merging the relevant rows in the table for $E(\{1, 2, 5\})$, we see that $\delta_R(1, b) = \{1, 2, 3, 5\}$.

After processing state 1, we add these new transitions to M', shown below





ε-Closure Table

State	ε-Closure
1	{1, 3}
2	{2}
3	{3}
4	{3, 4}
5	{5}

Handling State 2

1. Calculate the ϵ-closure of the state in M; call the result S. From the ϵ-closure table, we know that $E(\{2\}) = \{2\}$.

2. Calculate the transitions in M from each state in R on each input character.

   a Here we must collect all places we can get to when reading an **a** starting in any state in $\{2\}$. Since

   – $\delta(2,$ **a** $) = \{3, 4\}$

   we see then that $\delta_R(2,$ **a** $)$ includes at least $\{3, 4\}$.

   b Here we must collect all places we can get to when reading an **b** starting in any state in $\{2\}$. Since

   – $\delta(2,$ **b** $) = \{2\}$

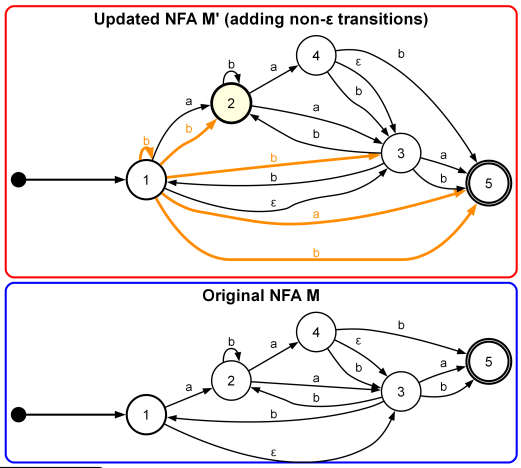
   we see then that $\delta_R(2,$ **b** $)$ includes at least $\{2\}$.

3. For each input character, calculate the ϵ-closure of the resulting set of states.

   a Merging the relevant rows in the table for $E(\{3, 4\})$, we see that $\delta_R(2, a) = \{3, 4\}$.

   b Merging the relevant rows in the table for $E(\{2\})$, we see that $\delta_R(2, b) = \{2\}$.

After processing state 2, we add these new transitions to M', shown below



ε-Closure Table

State	ε-Closure
1	{1, 3}
2	{2}
3	{3}
4	{3, 4}
5	{5}

Handling State 3

1. Calculate the ϵ-closure of the state in M; call the result S. From the ϵ-closure table, we know that $E(\{3\}) = \{3\}$.

2. Calculate the transitions in M from each state in R on each input character.

   a Here we must collect all places we can get to when reading an **a** starting in any state in $\{3\}$. Since

   – $\delta(3,$ **a** $) = \{5\}$

   we see then that $\delta_R(3,$ **a** $)$ includes at least $\{5\}$.

b Here we must collect all places we can get to when reading an **b** starting in any state in $\{3\}$. Since

- $\delta(3,$ **b** $) = \{1, 2, 5\}$

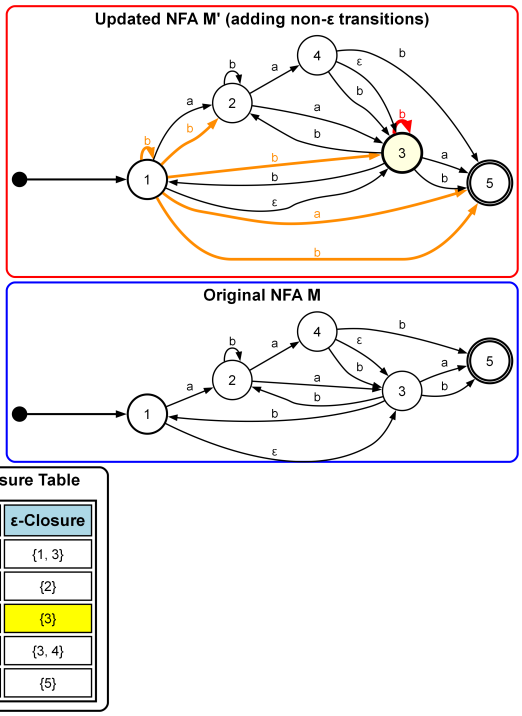
we see then that $\delta_R(3,$ **b** $)$ includes at least $\{1, 2, 5\}$.

3. For each input character, calculate the ϵ-closure of the resulting set of states.

a Merging the relevant rows in the table for $E(\{5\})$, we see that $\delta_R(3, a) = \{5\}$.

b Merging the relevant rows in the table for $E(\{1, 2, 5\})$, we see that $\delta_R(3, b) = \{1, 2, 3, 5\}$.

After processing state 3, we add these new transitions to M', shown below



State	ε-Closure
1	{1, 3}
2	{2}
3	{3}
4	{3, 4}
5	{5}

Handling State 4

1. Calculate the ϵ-closure of the state in M; call the result S. From the ϵ-closure table, we know that $E(\{4\}) = \{3, 4\}$.

2. Calculate the transitions in M from each state in R on each input character.

a Here we must collect all places we can get to when reading an **a** starting in any state in $\{3, 4\}$. Since

- $\delta(3,$ **a** $) = \{5\}$
- $\delta(4,$ **a** $) = \varnothing$

we see then that $\delta_R(4,$ **a** $)$ includes at least $\{5\}$.

b Here we must collect all places we can get to when reading an **b** starting in any state in $\{3, 4\}$. Since

- $\delta(3,$ **b** $) = \{1, 2, 5\}$
- $\delta(4,$ **b** $) = \{3, 5\}$

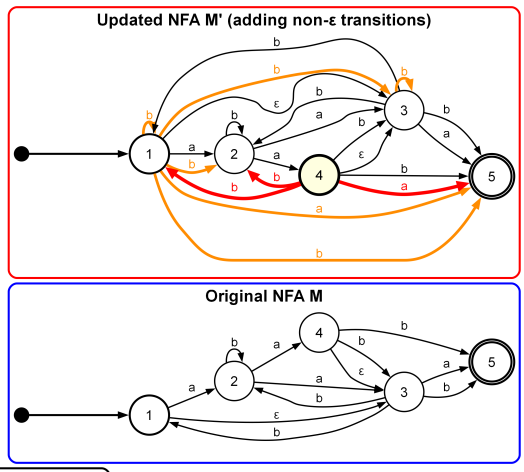
we see then that $\delta_R(4,$ **b** $)$ includes at least $\{1, 2, 3, 5\}$.

3. For each input character, calculate the ϵ-closure of the resulting set of states.

a Merging the relevant rows in the table for $E(\{5\})$, we see that $\delta_R(4, a) = \{5\}$.

b Merging the relevant rows in the table for $E(\{1, 2, 3, 5\})$, we see that $\delta_R(4, b) = \{1, 2, 3, 5\}$.

After processing state 4, we add these new transitions to M', shown below



ε-Closure Table

State	ε-Closure
1	{1, 3}
2	{2}
3	{3}
4	{3, 4}
5	{5}

Handling State 5

1. Calculate the ϵ-closure of the state in M; call the result S. From the ϵ-closure table, we know that $E(\{5\}) = \{5\}$.

2. Calculate the transitions in M from each state in R on each input character.

   a Here we must collect all places we can get to when reading an **a** starting in any state in $\{5\}$. Since

   – $\delta(5,$ **a** $) = \varnothing$

   we see then that $\delta_R(5,$ **a** $)$ includes at least $\{set()\}$.

   b Here we must collect all places we can get to when reading an **b** starting in any state in $\{5\}$. Since

   – $\delta(5,$ **b** $) = \varnothing$

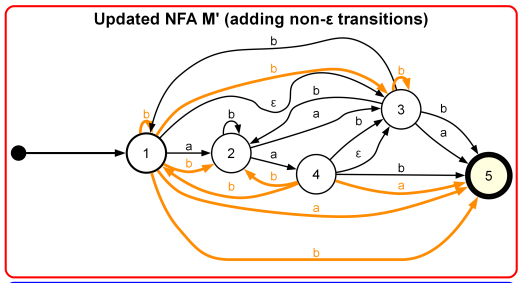
   we see then that $\delta_R(5,$ **b** $)$ includes at least $\{set()\}$.

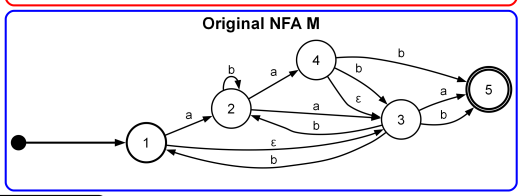
3. For each input character, calculate the ϵ-closure of the resulting set of states.

   a Merging the relevant rows in the table for $E(\{set()\})$, we see that $\delta_R(5, a) = \{set()\}$.

   b Merging the relevant rows in the table for $E(\{set()\})$, we see that $\delta_R(5, b) = \{set()\}$.

After processing state 5, we add these new transitions to M', shown below
We have now successfully removed all ϵ-transitions from M to produce M'.





ε-Closure Table

State	ε-Closure
1	{1, 3}
2	{2}
3	{3}
4	{3, 4}
5	{5}